

Air University



# Computer Networks (Lab)

Year 2025

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Academic Year: 2024 – 2028

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BS – Cyber Security – Fall – 24 (A)

Date of Submission: December 27, 2025

## **Assignment # 02**

### **TASK 1: MATCHING SCENARIOS TO TRANSMISSION MEDIA**

| Scenario                                                | Selected Medium | Rationale                                                                                                                                                                          |
|---------------------------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. PC → Switch (20m, same room, low cost)               | A) UTP (Cat6)   | Cat6 can support speeds of 10 Gbps up to 55 meters, making it an affordable choice for short connections in the same room.                                                         |
| 2. Cable near heavy electrical equipment (EMI expected) | B) STP (Cat6)   | Shielded twisted pair (STP) protects against electromagnetic interference (EMI) thanks to its metal shielding, which is important in noisy environments with motors and elevators. |
| 3. Two buildings 300m apart (high speed & reliability)  | D) Fiber Optic  | Fiber optic cables reliably transmit data over distances up to 300 meters for 10 Gbps and are immune to EMI, making them ideal for connecting buildings.                           |
| 4. Home TV/cable Internet line                          | C) Coaxial      | RG-6 coaxial cable is the standard for home cable TV and Internet connections.                                                                                                     |
| 5. Temporary classroom connection (no cables allowed)   | E) Wi-Fi        | The 802.11 wireless standard provides temporary, cable-free connectivity for classrooms, avoiding the need for physical connections.                                               |

### **TASK 2: TRANSMISSION MEDIA CHARACTERISTICS TABLE**

| Medium      | Guided/Unguided | EMI Resistance       | Typical Conductor |
|-------------|-----------------|----------------------|-------------------|
| UTP (Cat6)  | Guided          | Low                  | RJ45              |
| STP (Cat6)  | Guided          | High                 | RJ45              |
| Coaxial     | Guided          | High                 | BNC               |
| Fiber Optic | Guided          | High (immune to EMI) | SC, LC, ST, FC    |
| Wi-Fi       | Unguided        | Medium               | None (wireless)   |

**TASK 3: LAB LINK SELECTION WITH JUSTIFICATION**

| Link                                          | Selected Medium | Justification                                                                                                                                                                                   |
|-----------------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Switch → PC (25m)                             | A) UTP (Cat6)   | At 25 meters, Cat6 works well at speeds of 10 Gbps and is cost-effective for local area networks (LAN). The distance is within the 55-meter limit for high-speed use.                           |
| Switch → Switch (80m, near electrical wiring) | B) STP (Cat6)   | While 80 meters is longer than the 55-meter limit for high-speed Cat6, it is still acceptable for 1 Gbps over 100 meters. STP protects against EMI from nearby electrical wiring and equipment. |

**THE END**

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